Novelty Detection in Power Quality Signals with Surrogates: a Time-Frequency Technique

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Abstract - **In the new context of smart grids (SGs), new types of electric disturbances may occur in the electrical power systems (EPSs) current and voltage waveforms. The detection and the study of such disturbances can be useful in the operation and control of the EPSs. In order to identify any waveform abnormality or variation, novelty detectors might be used. This work presents a novelty detection technique for power quality signals based on surrogates and distances calculated in the time-frequency domain. Surrogates are signals created with the same magnitude of the Fourier spectrum from the original signal, but with the phase spectrum generated from a random uniform distribution. Tests using several cases of the most common power quality disturbances using synthetic signals were conducted. A comparison with an energy-based novelty detector currently present in the literature is performed, showing better performance results in terms of the receiver operating characteristic (ROC) curve parameters. Also, a real-life waveform signal is used to show the presented technique in action. The surrogate technique presented in this work can be useful in the definition of the novelties ground-truth for real power signal waveforms.**

Keywords – **surrogate, signal processing, novelty detection, power quality**

I. Introduction

Electric power systems (EPSs) have undergone a significant expansion in recent years [1]. This led to the need for new types of measurement, control, and data transmission instruments as a means of quality and stability maintenance. The modernization and interconnection of the grid through the insertion of new technological equipment is part of the establishment of smart grids (SGs) [2].

SGs allow new possibilities for a more flexible, sophisticated and versatile system, where new types of time-varying nonlinear loads and distributed power sources are present [3]. In this scenario, voltage and current signals may suffer new electrical disturbances, generating new power quality problems, for which EPSs must be prepared.

An important task is to identify any type of change or variation in the voltage and current waveforms, not just the disturbances already defined in the standards and literature (sags, swell, harmonics, transients, etc.) [4, 5], but any new event that may occur in the new context of SGs [6]. Such a task has been termed as "novelty detection". The concept of novelty detection in power quality signals the way it is approached here is relatively recent in the literature [7].

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There are some works approaching the novelty detection topic in the literature. In [8], an experimental evaluation of representative novelty detection methods is conducted. A comparative evaluation of four methods is carried out. The experimental results demonstrate that the *k*-NN novelty detection method exhibits competitive overall performance to the other methods in terms of the receiver operating characteristic (ROC) area under the curve (AUC) metric.

An automatic detector and classifier of voltage waveforms is proposed in [9]. It includes both the classification of the waveforms in multiple known classes and the detection of new waveforms (novelties) that are not available during the training stage. The classification method is based on the Support Vector Data Description (SVDD). The results obtained on both simulated and real-world data demonstrate the ability of the method to identify novelties and to classify known examples correctly.

The work described in [10] presents an approach for single and multiple power quality disturbance detection and classification using multidimensional analysis, higher-order statistics and a neuro-tree based classifier. It is able to detect deviations in the measured voltage waveform from the nominal one and classify 20 classes of single and multiple disturbances with a global efficiency upper to 97%.

An energy-based novelty detector is presented in [7]. This technique is used for comparison purposes in this paper. More details about this energy-based detector functionality are present in Section IV.

In the present work, a technique that uses surrogates adapted to detect novelties in power quality signals is presented. To do so, the detection of novelties is related to the stationary changes in the voltage and current signals. This technique can be useful in the evaluation of real voltage and current waveform signals in order to find the ground-truth of novelties present in them.

The surrogate technique is a time-frequency hypothesis testing tool, where an observation classification into stationary or non-stationary is pursued [11]. Here, we propose an application of this technique to power quality waveform signals, in order to identify the stationary changes related to novelties.

Simulation tests are conducted to characterize the performance of the technique. A real-life waveform signal is used to show the detection capacity of the presented technique.

The article is divided as follows. Section II shows the

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surrogate technique. Section III describes the implementation of the surrogate technique for novelty detection in power quality signals. Section IV briefly describes the energy-based novelty detector under comparison. In Section V, some detection results using both simulated and real-life signals are presented. Finally, Section VI states some conclusions.

II. The Surrogate Technique

A. Definitions

A discrete-time signal can be represented as a vector $\mathbf{x}(n)$, where the index n represents the sample position within the vector $\mathbf{x}(n)$. An N-length observation segment of this signal is a sequence represented by $\{\mathbf{x}_i(n), n = 1, \ldots, N\}$. If I observations from the signal $\mathbf{x}(n)$ with length N are considered, we can denote them by $\{\mathbf{x}_i(n), n = 1, \ldots, N; i = 1, \ldots, I\}$.

Each observation $\mathbf{x}_i(n)$ can be considered as a realization of a discrete-time stochastic process $X(n,i)$ related to the original signal $\mathbf{x}(n)$. Thus, this stochastic process and its realizations can be defined as:

$$\begin{aligned} &\text{Process: } \{X(n,i) : n,i \in \mathbb{N}\} \\ &\text{Realizations: } X(n,i) = \mathbf{x}_i(n)\,, n = 1,\ldots,N; i = 1,\ldots,I. \end{aligned} \tag{1}$$

Although the concept of stationary processes is uniquely defined in theory, it is interpreted in different ways in practical terms. While in the statistics field stationarity is defined as the invariance of statistical properties over time, it can also be applied to deterministic signals, whose spectral characteristics are invariant over time. In this work, novelty detection is associated with the change of stationarity within an observation. In other words, the change of the spectral characteristics.

B. Surrogates

The objective of the surrogate technique is to analyze a realization in a given observation scale and classify it as belonging to a stationary or non-stationary process, considering it as the only realization of the process.

Surrogates can be understood as a stationary reference for a realization $\mathbf{x}_i(n)$. For the same marginal Fourier spectrum, non-stationary processes are different from stationary processes by a structured organization over time. A set of J surrogates can be computed for an $\mathbf{x}_i(n)$ realization in such a way that each surrogate has the same Power Spectral Density (PSD) as the original signal, while being stationary. In practice, this can be achieved by destroying the organized phase structure that controls non-stationarity, if any.

To create a surrogate $\mathbf{s}_j(n)$, the realization $\mathbf{x}_i(n)$ is first Fourier-transformed. The magnitude of the Fourier spectrum is kept unchanged, while the phase is replaced by a random sequence. On this modified spectrum, the Inverse Fourier Transform is applied, from which the surrogate $\mathbf{s}_j(n)$ is obtained. More surrogates can be created, as many as desired, just performing new random phase replacements and applying the Inverse Fourier Transform, resulting in a set with J surrogates. Creating surrogates as described above forms a stationary reference set for the given realization $\mathbf{x}_i(n)$ [11].

The next paragraph shows the math equations related to the described procedure.

Consider that the realization $\mathbf{x}_i(n)$ has a Discrete Fourier Transform (DFT) $X_i(k)$ with bins indexed by k with a magnitude $A_i(k)$ and phase $\Phi_i(k)$ such that

$$\begin{aligned} X_i(k) &= A_i(k)e^{j\Phi_i(k)} = A_i(k)\angle\Phi_i(k) \\ \mathbf{x}_i(n) &= \frac{1}{N}\sum_k X_i(k)e^{\frac{j2\pi nk}{N}} = \mathcal{F}^{-1}\left(X_i(k)\right), \end{aligned} \tag{2}$$

where $\mathcal{F}^{-1}(\cdot)$ stands for the inverse Fourier transform. A surrogate $\mathbf{s}_j(n)$ is created from $X_i(k)$ replacing $\Phi_i(k)$ with an independent, identically distributed (i.i.d.) phase sequence drawn from a uniform distribution over $[-\pi, \pi]$, named $\Psi_j(k)$:

$$\mathbf{s}_j(n) = \frac{1}{N}\sum_k A_i(k)e^{j\Psi_j(k)}e^{\frac{j2\pi nk}{N}} = \mathcal{F}^{-1}\left(A_i(k)e^{j\Psi_j(k)}\right). \tag{3}$$

C. Hermite windows

Hermite windows are an important element in the stationarity estimation technique for novelty detection. They are used for windowing the frames and the surrogates. The Hermite windows are a family of functions $h_q(t)$, where q is the window order. The qth-order Hermite window can be calculated as follows:

$$h_q(t) = \left((t-\mathcal{D})^q\, g(t)\right) \Big/ \sqrt{\pi^{\frac{1}{2}} 2^q q!}, \tag{4}$$

where operator $\mathcal{D}$ is the derivative operator $\frac{d(\cdot)}{dt}$ and $g = e^{\frac{-t^2}{2}}$. Fig. 7 shows three Hermite windows, of order 1, 2 and 3.

In practice, a set of windows limited to the Qth-order is used. The choice of Hermite windows is because they are orthonormal and maximally concentrated in the time-frequency domain [12].

D. Hermite spectrograms

A Hermite spectrogram $S_{x_i}^{(h_q)}(t,f)$ can be seen as a convolution of the frame signal $x_i(t)$ with a window formed by the product between the Hermite window $h_q(t)$ and the Fourier complex basis [11]. In other words, it can be considered a short-time Fourier transform (STFT) spectrogram with a Hermite window:

$$S_{x_i}^{(h_q)}(t,f) = \left|\int x(\tau)h_q(\tau - t)e^{-j2\pi f\tau}d\tau\right|^2. \tag{5}$$

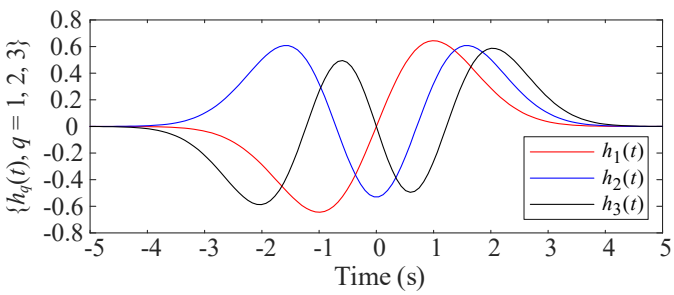

Figure 7. Hermite windows $h_1(t)$, $h_2(t)$ and $h_3(t)$.

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In practice, Hermite spectrograms are not calculated in all samples of the frame, meaning that the time resolution is not the sample period T. Instead, it is calculated at D instants of time, each instant represented by t_d, with $d = 1, \ldots, D$. The interval of calculation $(t_{d+1} - t_d)$ is usually half of the time width of Hermite window. The frequency is discretized taking the discrete values of f_d. Thus, the discrete-time Hermite spectrograms are calculated by replacing the integral in (5) by a summation for each window $\mathbf{h}_q(n)$, as in (6), where $\mathbf{h}_q(n)$ is the discrete-time Hermite window version.

$$S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d) = \left| \sum_{n=1}^{N} \mathbf{x}_i(n) \mathbf{h}_q(n-\tau) e^{-\frac{j2\pi f_d n}{N}} \right|^2, \quad \tau \in \{t_d, d = 1, \ldots, D\}, q = 1, \ldots, Q, \tag{6}$$

where $\mathbf{h}_q(n-\tau)$ corresponds to the qth-order discrete-time Hermite window shifted by τ samples, and $e^{-\frac{j2\pi f_d n}{N}}$ are the Fourier basis for the analyzed frame.

After calculating the Q Hermite spectrograms, the result are Q matrices, where each column represents a "local spectrum" calculated at each t_d instant.

E. Multitaper spectrograms

With Q Hermite spectrograms, an average of all Q matrices is calculated, element by element, results in a new average matrix, named multitaper spectrogram:

$$S_{\mathbf{x}_i,Q}(t_d, f_d) = \left\langle S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d) \right\rangle_q := \frac{1}{Q} \sum_{q=1}^{Q} S_{\mathbf{x}_i}^{(\mathbf{h}_q)}(t_d, f_d). \tag{7}$$

The multitaper spectrogram is also a matrix of the same order as the Hermite spectrograms. Its columns represent local (average) spectra. After calculating the multitaper spectrogram, all of its columns are averaged over the all t_d instants, resulting in the global (average) spectrum:

$$\left\langle S_{\mathbf{x}_i,Q}(t_d, f_d) \right\rangle_d := \tfrac{1}{D} \sum_{d=1}^{D} S_{\mathbf{x}_i,Q}(t_d, f_d). \tag{8}$$

F. Distance test

The global spectrum $\left\langle S_{\mathbf{x}_i,Q}(t_d, f_d) \right\rangle_d$ is a vector whose elements are mean squared amplitudes for each frequency of the Fourier spectrum. Distance quantities can be calculated to compare each local spectrum $S_{\mathbf{x}_i,Q}(t_d, f_d)$ and the global spectrum. Thus, the comparison between the D local spectra and the global spectrum form a set of distances $\{c_d^{(\mathbf{x}_i)}, d = 1, \ldots, D\}$, with D elements calculated along all frequencies f_d, defined as:

$$\left\{ c_d^{(\mathbf{x}_i)} := L\left(S_{\mathbf{x}_i,Q}(t_d, f_d), \left\langle S_{\mathbf{x}_i,Q}(t_d, f_d) \right\rangle_d \right), d = 1, \ldots, D \right\}, \tag{9}$$

where the distance $L(\cdot,\cdot)$ is composed of two distances, the Kullback-Leibler divergence and the logs-spectral deviation [13].

The calculations of this set of distances is based on the fact that if there is any variation in the spectrum of $\mathbf{x}_i(n)$ (non-stationarity), the variance of these D distances tends to be a greater value than when $\mathbf{x}_i(n)$ is stationary.

After calculating the D distances, for the frame $\mathbf{x}_i(n)$, we can calculate the variance of the set $\left\{ c_d^{(\mathbf{x}_i)}, d = 1, \ldots, D \right\}$ as follows:

$$\Theta_{\mathbf{x}_i} = \operatorname{var}\left(c_d^{(\mathbf{x}_i)} \right)_{d=1,\ldots,D}, \tag{10}$$

where $\operatorname{var}(\cdot)$ is the variance operator.

G. Non-stationarity detection

After the set of J surrogates is created as described in (3), the same steps described from Subsections D to F are performed for each surrogate $\mathbf{s}_j(n)$. Therefore, this results in J multitaper spectrograms, J global spectra and $J \times D$ local spectra. For each surrogate $\mathbf{s}_j(n)$, a set of distances $\{c_d^{(\mathbf{s}_j)}, d = 1, \ldots, D\}$ is calculated as shown in (9), totalizing J distance sets with D distances in each set. As in (10), the variance is calculated for each set, resulting in a set of J variances:

$$\left\{ \Theta_{\mathbf{s}_j} = \operatorname{var}\left(c_d^{(\mathbf{s}_j)} \right)_{d=1,\ldots,D}, j = 1, \ldots, J \right\} \tag{11}$$

The distribution of the set $\left\{ \Theta_{\mathbf{s}_j}, j = 1, \ldots, J \right\}$ allows the determination of a threshold γ above which the null hypothesis of stationarity is rejected (the observation is non-stationary). The median value of $\left\{ \Theta_{\mathbf{s}_j}, j = 1, \ldots, J \right\}$ can be used as the threshold γ, which is applied to $\Theta_{\mathbf{x}_i}$ taking the form of the following hypothesis test:

$$d(\mathbf{x}_i) = \begin{cases} 1, \text{ if } \Theta_{\mathbf{x}_i} \geq \gamma : \text{"non-stationarity"}; \\ 0, \text{ if } \Theta_{\mathbf{x}_i} < \gamma : \text{"stationarity"}. \end{cases}, \tag{12}$$

where $d(\mathbf{x}_i)$ is the decision function for the frame $\mathbf{x}_i(n)$. If $d(\mathbf{x}_i) = 1$ one can assume that a novelty was detected, otherwise, no novelty was detected.

III. Surrogate Technique Applied to Novelty Detection in Power Quality Signals

In order to apply the surrogate technique to detect novelties in power quality signals, it is necessary to define the length of the frames xi(n) to segment the signal x(n). The same length used in [7] was adopted, that is, 4 cycles of fundamental frequency. A sampling rate of 128 samples per cycle was used. So, the length of each frame is N = 512 samples. Considering the EPS fundamental frequency equal to 60 Hz, the resultant sampling rate is F_s = 7.68 kHz and the sampling period is approximately T = 130 μs.

A Hermite window width of one fundamental cycle was adopted, and the spectrogram time resolution $(t_{d+1} - t_d)$ was half the width of the Hermite window, that is, half a cycle. Thus, the Hermite spectrum is calculated in 7 instants of time within a frame (t_1, t_2, …, t_7). The maximum Hermite window order of Q = 4 was adopted. All implementation and test simulations for this paper were performed in MATLAB® environment.

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Fig. 8 shows a representation of a sinusoidal frame $\mathbf{x}_i(n)$ and the $\mathbf{h}_1(n)$ window being shifted over time. The same procedure is done with the windows $\mathbf{h}_2(n)$, $\mathbf{h}_3(n)$, and $\mathbf{h}_4(n)$ before the calculation of the Hermite spectrograms, one for each window.

Fig. 9 shows the $\mathbf{h}_1(n)$-Hermite spectrogram related to the frame $\mathbf{x}_i(n)$ from Fig. 3. Each spectrogram has the representations of the Hermite window in the background (red line). The frequency axis is limited to 1500 Hz for better visualization. Each Hermite spectrogram is a 7-column matrix whose columns correspond to the $D = 7$ local spectra. Their lines correspond to the Fourier frequencies.

After calculating the Hermite spectrograms, the multitaper spectrogram is calculated. It is done by averaging the four Hermite spectrograms, element by element, as defined in (7). The next step is the calculation of the global spectrum, which is obtained by averaging all the columns of the multitaper spectrogram along the 7 columns.

A representation of the multitaper spectrogram can be seen in Fig. 10 filled in orange. The global spectrum is filled in red, which is the time average of all local spectra, and it is positioned at the beginning of the frame for better visualization.

Once the multitaper spectrogram, the local and global spectra are calculated, the distances between all 7 local spectra and the global spectrum over all frequencies f_d are calculated according to (9). There are 7 distances per frame, forming the set $\{c_1^{(\mathbf{x}_i)}, c_2^{(\mathbf{x}_i)}, \ldots, c_7^{(\mathbf{x}_i)}\}$, which has the variance calculated as in (10).

The surrogate methodology consists of performing the same

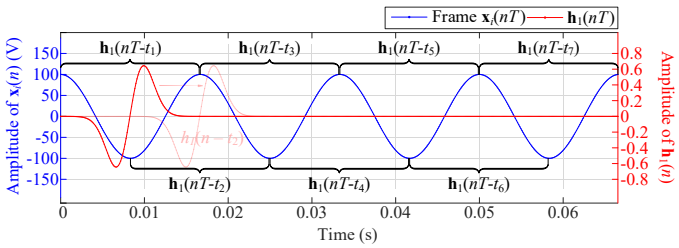

Figure 8. Windowing process of frame $\mathbf{x}_i(n)$ with the window $\mathbf{h}_1(n)$.

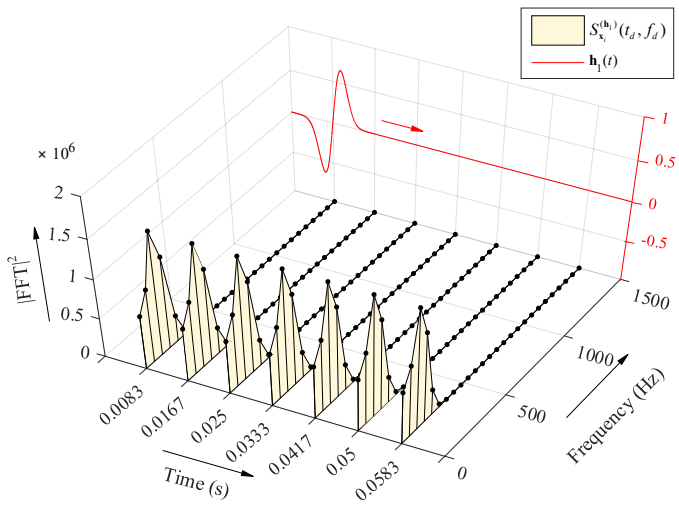

Figure 9. Hermite spectrogram of $\mathbf{x}_i(n)$ from Fig. 3 using $\mathbf{h}_1(n)$.

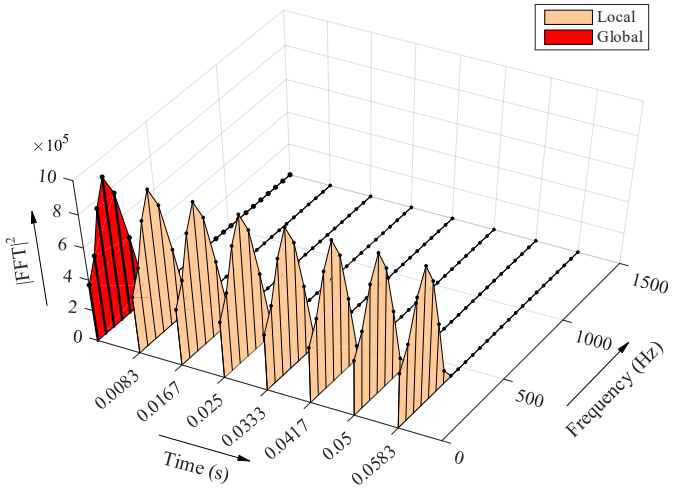

Figure 10. Multitaper spectrogram for the frame of $\mathbf{x}_i(n)$ from Fig. 3 along with the global spectrum.

calculations of all Hermite spectrograms and distances for each surrogate created for each frame $\mathbf{x}_i(n)$. Thus, the next step is the creation of surrogates. A surrogate is created by using the magnitude of the Fourier Transform and randomly selecting phase components as described in the Subsection *B*.

IV. Energy-Based Novelty Detector [7]

In this paper, the surrogate technique is compared to the novelty detector proposed by [7], which is an energy-based detector. The current section is intended to briefly explain how this energy-based detector works.

This detector also requires the fixed segmentation into frames with a length of four fundamental cycles. The detection is built on comparing the current frame with a reference one according to some metrics. The work shows that the energy difference between the compared frames is the best metric among the tested metrics. A novelty detection is flagged when a signal frame is different from the reference one. The reference frame is the last detected novelty frame.

The detector structure is composed of two main branches. The upper branch is composed of a high-pass filter and the energy difference evaluation followed by a thresholding process. The purpose of the high-pass filter is to attenuate the effect of the 60 Hz fundamental component (highest-energy component) and low-frequency harmonics. In this way, this branch is responsible to detect novelties related to higher frequency disturbances.

The lower branch of the detector improves the performance of the detector. It is composed by a DFT centered in 60 Hz calculation, and a comparison of the magnitude and phase for each frame. There are two main objectives for this lower branch: detect low frequency disturbances, such as flicker and sub-harmonics, and signalize if the signal presents frequency variations.

V. Results

Reference [11] shows that a set of $J = 50$ surrogates has good representation for a stationary reference. This is the number of the surrogates used in this work. After the creation of 50

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surrogates per frame, for each surrogate the multitaper spectrogram and the distance set are calculated. Then, the stationary hypothesis test from (12) is applied to detect novelties.

A. Simulation results

Fig. 11 shows a part of a synthetic signal composed of five frames. The highlighted frames in the upper plot are the ones that contain novelty. The first frame is a novelty frame, as there is a stationary change due to the transient present during the second cycle. The second frame does not present any stationary change since the spectral content remains the same during the frame. In the third frame, there is the insertion of the twentieth harmonic in the middle of the frame, representing a stationary change, and therefore, a novelty. The fourth frame keeps the spectrum unchangeable showing no novelty. Finally, in the last frame, the twentieth harmonic is removed, which represents a stationary change, which makes the frame a novelty frame.

The middle plot shows 50 surrogates for each frame. In the bottom plot the frames that the surrogate technique detected as novelty frames are highlighted. The threshold γ is shown in black line while the tested quantity $\Theta_{\mathbf{x}_i}$ is in blue line. One can see that the frames were classified correctly, since the blue line is above the black line only in the novelty frames.

The example shows only a qualitative result. Further tests with several types of power quality disturbances using longer signals have been performed to prove the effectiveness of the surrogate technique. A comparison with the technique proposed in [7] is also carried out. To compare the novelty detectors, the following disturbances were tested: oscillatory transients, sags, swells, time-varying harmonics, notches, spikes, interruptions, phase shifts, DC offset, and time-varying interharmonics.

For each disturbance described above, 200 random realizations were generated, where each realization consisted of 10 frames, totaling 2000 frames for classification. The generated signals were normalized by the fundamental component. In addition, in order to test the robustness of the detectors with the presence of noise, white Gaussian noise with SNR of 30 dB was added to the signals.

For performance analysis and comparison of the detectors, the Receiver Operating Characteristic (ROC) curve is used [14]. Each detector works as a frame classifier in two classes: novelty frames and non-novelty frames. The ROC curve applied to the detectors relates the detector's ability to classify a novelty frame when it actually occurred (probability of detection P_D), with the

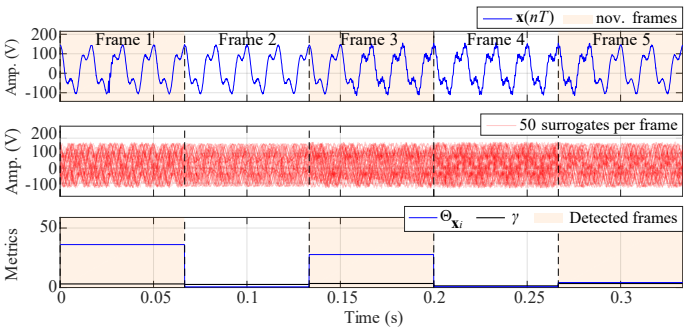

Figure 11. Novelty detection with surrogates. Top plot: $\mathbf{x}(n)$. Middle plot: surrogates. Bottom plot: novelty detection.

frequency that the detector classifies novelty frames when it did not occur (false alarm probability P_{FA}).

Fig. 12 shows two ROC curves related to novelty detection tests with oscillatory transient disturbances. The blue curve and shade correspond the performance related to the detector based on the surrogate technique. The red curve and fill represent the classification performance related to the technique based on [7]. The points highlighted on each curve are the detection optimal threshold values for both techniques, calculated according to [15]. The closer the optimum point is to the upper left corner the better the estimator. It is possible to see from Fig. 12 that the surrogate ROC curve shows better performance for this type of disturbance.

For better visualization and interpretation of results, instead of plotting all ROC curves for all disturbances for both detectors, some metrics were extracted from the ROC curves. They are P_D and P_{FA} at the optimum point, and the area under the curve (AUC) for each disturbance. The performance evaluation criteria for the above metrics are the closer to 1 is P_D, the closer to 0 is P_{FA} and the closer to 1 is AUC the better the detector.

Table I shows the distribution of P_D, P_{FA} and AUC for all disturbances for both novelty detectors. One can see that the surrogate-based P_D performed better in all tested disturbances, because all of them are closer to 1 than the P_D resultant from [7].

For P_{FA} values, the surrogate-based detector performed better in most cases, except for tests with interharmonics. This probably

TABLE I. PERFORMANCE COMPARISON

Disturb.	P_D		P_{FA}		AUC	
	Surr.	[7]	Surr.	[7]	Surr.	[7]
Transients	0.9921	0.8783	0.0034	0.0299	0.9927	0.9808
Sags	0.9450	0.9378	0.0137	0.0814	0.9881	0.9720
Swells	0.9843	0.9698	0	0.0266	0.9891	0.9765
Harmonics	0.9950	0.8375	0.0013	0.0018	≈1	0.9684
Notches	0.9984	0.8439	0	0.1651	≈1	0.9103
Spikes	0.9634	0.8259	0.0080	0.1905	0.9983	0.8628
Interruption	0.9292	0.9305	0.0649	0.3169	0.9727	0.8380
Phase shift	0.9946	0.9405	0	0.2530	0.9977	0.9144
DC offset	0.9767	0.9396	0.1313	0.2091	0.9425	0.9267
Interharmonics	0.9300	0.8350	0.4539	0.0381	0.7817	0.9760

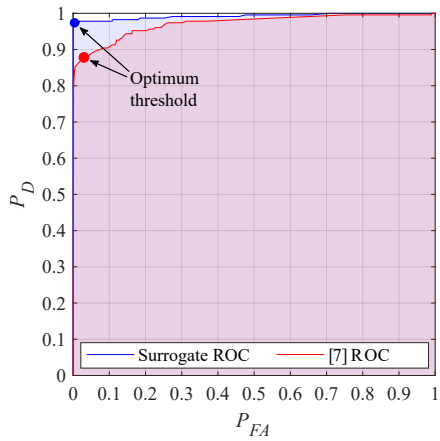

Figure 12. ROC curves for surrogate technique (blue) and the approach present in [7] (red) considering oscillatory transient disturbances.

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happen because of the adopted frame size (four fundamental cycles), which limits the resolution of the fast Fourier transform (FFT) in 15 Hz. Non-multiple interharmonics of this resolution may not be detected. During the tests, any value of interharmonics within the Nyquist range was allowed.

For AUC values, the detector based on surrogates performed better since more often the values approach 1. The exception is again the interharmonics.

All simulation signals were generated using MATLAB®.

B. *Real signal novelty detection*

In Fig. 7, the surrogate technique applied to a single-phase current waveform signal acquired from a wind generator [16].

In the top plot of Fig. 7, a current waveform with 84 four-cycle frames is shown. The waveform duration is 5.6 s. The bottom plot shows a zoom from $t = 2.5$ s to $t = 3.5$ s, where it is possible to see an abrupt event in the frame just after $t = 3$ s.

The color scheme adopted here is the same of the top plot of Fig. 11, that is, the novelty frames are highlighted. We can see that the technique was able to detect all variations present in the signal, including the abrupt distortion magnified in the bottom plot.

An important consideration on the detection sensitivity of the test (12) is that it can be controlled through a multiplicative factor applied to γ. The greater this factor, the less sensitive the technique becomes and, therefore, the more susceptible to loss of disorders. On the other hand, the lower the multiplicative factor, the more sensitive the technique behaves and the more subtle events are detected.

VI. Conclusion

A time-frequency technique based on surrogates for novelty detection in power quality signal waveforms was presented in this work. Tests involving several disturbances were performed using simulation signals, and a comparison with another detector currently present in the literature was made.

The results show that the detector based on surrogates performed better in almost all cases, in terms of the ROC curve parameters. Also, the technique was applied to a real current signal.

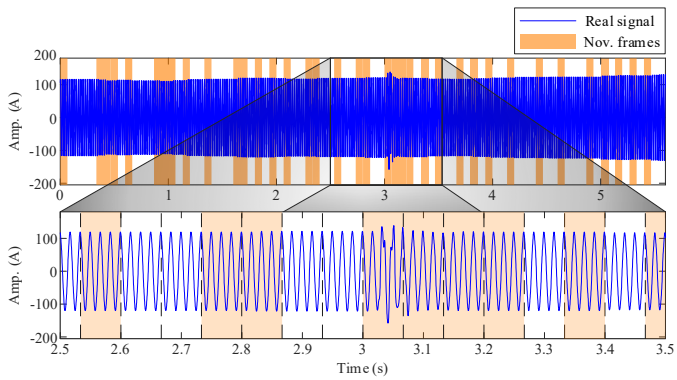

Figure 7. Novelty detection applied to a real current waveform signal. Top plot: 84 frame length waveform. Bottom plot: magnification.

Future work consists of improving the Fourier frequency resolution. A possibility would be to increase the frame length. However, this could decrease the time resolution.

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